

Lecture 6: Risk, Safety, and Accidents

HUM 4441: Engineering Ethics

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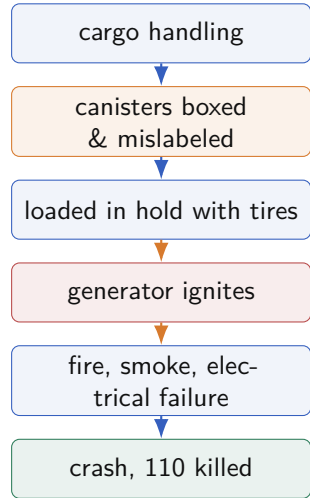
ValuJet Flight 592

On a sunny afternoon in May 1996, ValuJet Flight 592 left Miami for Atlanta and, within minutes, filled with smoke. Improperly handled oxygen generators had been packed with other cargo in the forward hold, where a fire started beneath the cockpit. The pilots tried to turn back, but the fire spread, the controls failed, and the aircraft crashed into the Everglades. All 110 people aboard were killed.

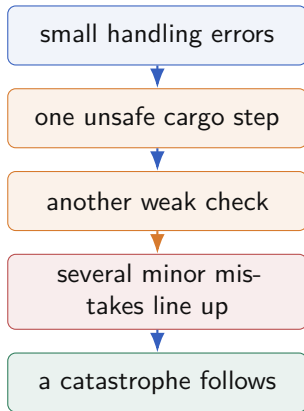
Framing question

What responsibility does an engineer have when a disaster is produced by a chain of handling, packing, and design decisions rather than by one obvious broken part?

Fleddermann, Ch. 5, p. 75



Why this crash matters



The first lesson

This was not just a pilot mistake. It was a chain failure in which maintenance, packing, shipping, loading, and aircraft design each left room for the next error to matter more.

Why the chapter starts here

Safety is not only about preventing one failure. It is about designing and operating systems so that small mistakes do not combine into disaster.

Fleddermann, Ch. 5, p. 75

The question to keep in view

Risk & safety

How safe is safe enough when the cost of failure is a loss of life?

Design duty

Which failures should an engineer foresee before the product is used?

Accident type

When do many small mistakes become a systemic failure?

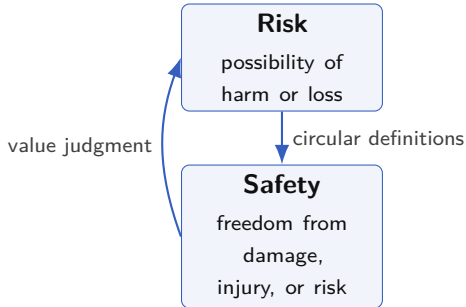
Framing question

If a design is only safe when everyone else behaves perfectly, is the design actually safe?



Fleddermann, Ch. 5, pp. 75–80

Risk and safety are judgments, not absolutes



Why the terms feel vague

A product can be called safe by one person and unsafe by another. The words are precise in a dictionary, but the judgment depends on experience, exposure, and who bears the harm.

Main point

Nothing engineering produces is 100% risk free. The real task is to make the design as safe as reasonably possible.

Fleddermann, Ch. 5, pp. 75–76

People judge risk by several cues

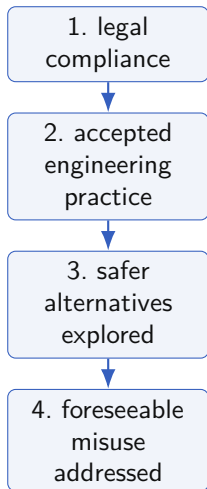
Factor	Risk feels <i>lower</i> when ...	Risk feels <i>higher</i> when ...
Voluntary vs. involuntary	people choose the exposure	it is imposed on them
Short-term vs. long-term	harm is temporary	harm is permanent or lasting
Expected probability	the chance is tiny	the chance is frequent or certain
Reversible effects	the damage can be undone	the damage cannot be reversed
Threshold levels	danger appears only at high exposure	danger exists even at low exposure
Delayed vs. immediate	harm is far in the future	harm is immediate

Engagement

A project creates a low-probability but irreversible harm for neighbours who did not choose to be near it. Which factor should matter most: probability, reversibility, or involuntariness?

Fleddermann, Ch. 5, p. 76

Four criteria for a safe design



Legal

minimum requirement

Practice

must match what the profession accepts as safe

Alternatives

compare other designs before choosing

Misuse

design for predictable misuse, not only ideal use

Why it gets harder

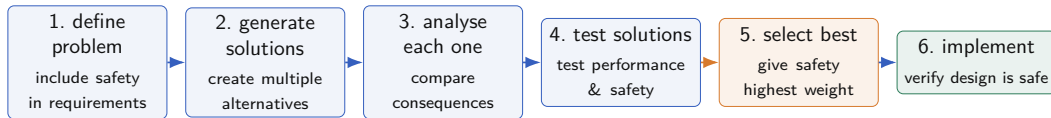
Each step asks for more judgment, more creativity, and more responsibility. A warning label is not a substitute for a safer design.

Engagement

If a design meets the law but a plausible misuse can still cause serious injury, what should change first: the label, the instructions, or the product itself?

Fleddermann, Ch. 5, pp. 77–78

Safety belongs in every step of the design process



Practical rule

Safety enters at the start, but it weighs most heavily when the final trade-offs are made. If the last choice is between cheaper and safer, safety carries the greater weight.

CS connection

The same logic applies to APIs and deployments: safe defaults belong in the design, not only in the help page or post-release patch.

Engagement

At which step is it hardest to recover from a late safety warning: defining the problem, testing, or selecting the final design?

Risk-Benefit Analysis and Environmental Racism



Homes built around a former toxic landfill site.

What the method does

Risk-benefit analysis assigns dollar values to risks and benefits and looks for the most favourable ratio. The numbers are often uncertain approximations.

Environmental racism

The disproportionate placement of hazardous facilities (landfills, chemical plants, waste incinerators) near communities of colour. These communities bear the health risks while wealthier areas receive the economic benefits.

When it occurs

Environmental racism arises when site selection prioritises cheap land and minimal political resistance over equitable risk distribution. It happens during zoning, permitting, and corporate location decisions.

Engagement

If a project saves money only by moving the hazard onto one neighbourhood, what would have to be true before you would call the decision ethical?

Carry these three checks into the accident cases

Procedural

A bad choice or a missed procedure. Training, supervision, and enforcement can reduce it.

Engineered

A design flaw or a device that fails under real conditions. Testing and redesign are the main defences.

Systemic

Many small failures interact inside a complex system. No single fix is enough.

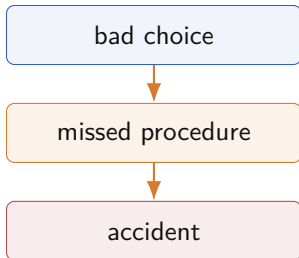
Why ValuJet returns later

The hardest failures are the ones that look harmless one step at a time. ValuJet is the chapter's standard example of that pattern.

Engagement

Which accident type is hardest to control in a cloud service or app deployment: one bad step, one bad design choice, or a long chain of small mistakes?

Procedural accidents



What it means

Procedural accidents come from someone making a bad choice or failing to follow established procedures. In aviation this is often called “*pilot error*”, but the same idea applies to maintenance, inspection, and engineering sign-off.

Typical fixes

training, supervision, new rules, closer scrutiny

Engineering duty

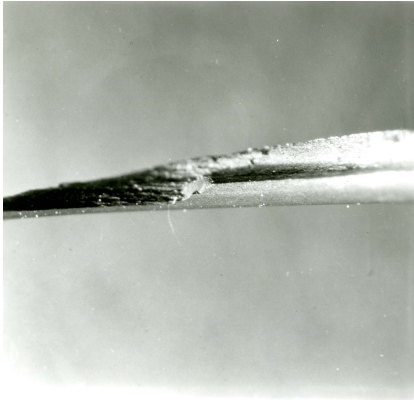
do not sign off on drawings or procedures you have not checked

Engagement

Which would stop a procedural accident faster in your own work: a better checklist, better training, or better oversight? Pick one and defend it.

Fleddermann, Ch. 5, pp. 80–81

Engineered accidents



A fatigue crack in a failed spring section.

What it means

Engineered accidents come from flaws in the design: a material fails, a device does not perform as expected, or it fails when operating conditions change. Microcracks in aircraft turbine blades are a classic example.

Design lesson

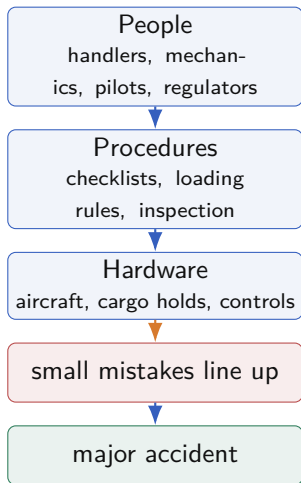
Anticipate the edge cases during design and test the product for safety, not only for specification compliance.

CS connection

A bug that appears only under rare load is still an engineered failure. Rare does not mean harmless.

Fleddermann, Ch. 5, p. 80

Systemic accidents



What makes systemic accidents different

They happen in complex technologies and complex organisations. One minor mistake is often not enough; disaster appears when several small errors converge.

Why they are hard to prevent

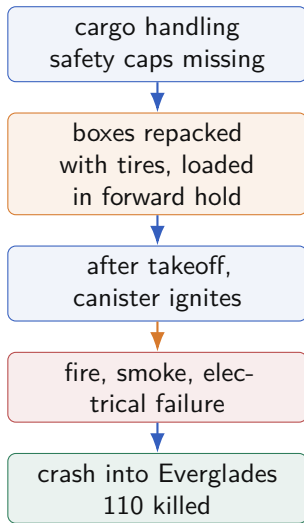
No single person owns the whole chain. That makes design of procedures, manuals, training, and interfaces part of the engineer's responsibility.

Preview

ValuJet 592 is the chapter's clearest example: one unsafe cargo chain became a disaster because the system let many small mistakes survive together.

Fleddermann, Ch. 5, pp. 80–81

The Crash of ValuJet Flight 592



Classification

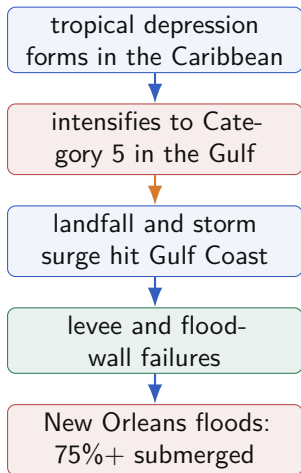
ValuJet 592 is a systemic accident. The crash came from the convergence of packaging, loading, acceptance, and design weaknesses.

What should have stopped the chain

Safety caps, proper packing, cargo rejection by the ramp agent, and smoke/heat detectors in the hold were all possible break points.

Fleddermann, Ch. 5, pp. 82–84

Hurricane Katrina



Type

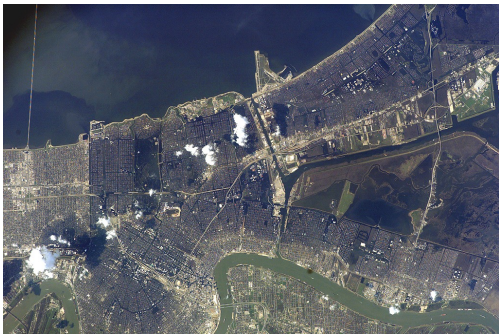
Systemic, with serious engineered failures in the levee and floodwall system.

ASCE's central lesson

The original design judgements were not adequate for critical life-safety structures. Calculations omitted key loading and soil-strength issues, and the design sat too close to the margin. A protection system must be designed for the hazard that can actually occur, not only for the historical record.

Fleddermann, Ch. 5, pp. 80–82

Hurricane Katrina: risk was unevenly distributed



NASA satellite view of New Orleans after Katrina.
Dark areas are flooded; lighter areas remained dry.

What the case adds

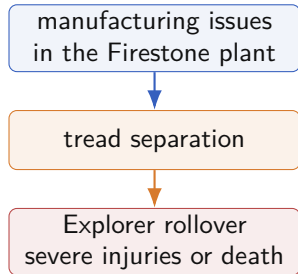
The benefits of the city, port, and development were not matched by an equal distribution of flood risk. The people behind the levees bore the most severe danger.

Engineering responsibility

Codes of ethics require engineers to consider how risks are distributed, not only whether the design meets the specification.

Fleddermann, Ch. 5, pp. 81–82

Firestone Tires / Ford Explorer



Type

Engineered failure sharpened by systemic interaction between tire quality, vehicle behaviour, and corporate response.

Trigger

field reports from hot climates and rollover crashes

Weak point

adhesives past shelf life; tire bubbles punctured during production

Response

recall, public dispute, Ford break with Firestone, Ford-owned recall

Lesson

Parts that look acceptable alone can fail badly when combined. Engineers must test the system, not just the component.

Fleddermann, Ch. 5, pp. 84–86

The Collapse of the Hyatt Regency Kansas City Walkways



Type

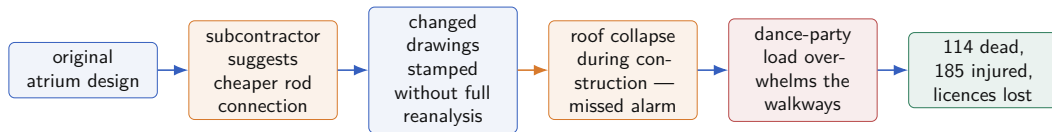
Engineered and procedural. A design change altered the load path, and the change was approved too casually.

Lesson

A change to a structural drawing is not clerical work. It is new engineering and must be checked as carefully as the original design.

Fleddermann, Ch. 5, pp. 86–87; Figs. 5.1 and 5.2

Hyatt Regency: how the failure chain grew



Case-analysis prompt

What should Gillum-Colaco have required before the change was approved: a new calculation, an on-site review, or a stop-work order until the load path was rechecked?

Why the case matters

The original design was already marginal. The change made the supporting nuts carry twice the intended load.

Fleddermann, Ch. 5, pp. 86–87

The Ford Crown Victoria Police Interceptor



A Ford Crown Victoria police car in service.

Type

Engineered, with procedural and organisational gaps. The design met federal standards, but police use made the rear-impact risk much higher than the civilian version suggested.

Problem

tank in rear crush zone near suspension bolts

Response

bulletin, shields, then fire suppression added to later models

Lesson

compliance is not enough when the operating context changes the hazard

Engineering lesson

Acceptable practice has to match the real use case, not only the legal minimum.

Fleddermann, Ch. 5, pp. 87–89

The Failure of the Teton Dam



Fig. 5.3. Downstream flooding after the dam collapsed in June 1976, showing how much water escaped once the first weak point gave way.

Type

Engineered and procedural. The dam was a familiar earthfill structure, but the fractured canyon rock, the grout curtain, and the monitoring plan did not provide enough defence.

- **Weak point:** heavily fractured rock beneath and beside the dam.
- **Failure path:** leakage, erosion, hydraulic fracturing, and collapse.
- **Lesson:** do not overtrust a single barrier; design the monitoring and leakage handling the site will actually need.

Fleddermann, Ch. 5, pp. 89–91; Fig. 5.3

The DC-10: design choices that made failure worse

1970: prototype
pressure test, cargo
door blows out



1972: near-miss in
Windsor, hydraulic lines
severed, safe landing



1974: Paris crash, door
failure, total loss of
control, 346 killed



1979: Chicago, en-
gine separation,
all on board killed

Type

Engineered and procedural, with systemic pressure from schedule and corporate culture.

Latch choice

electric latch was lighter, but the failure mode was catastrophic at altitude

Airframe choice

only three redundant hydraulic systems vs. four used by Boeing and Lockheed; lines routed through the floor, not the ceiling

Engineering lesson

A design can look efficient and still be unsafe if its failure mode is too violent. Conservative practice from the rest of the industry matters.

Fleddermann, Ch. 5, pp. 91–93

The DC-10: what corporate practice added

The inspection problem

After the 1972 near-miss, McDonnell Douglas agreed to modify the door latches. Three inspectors stamped the records even though the work had not been done.

The culture problem

The company did not push the design far enough from the older DC-8/DC-9 structure, and economic pressure encouraged a cost-conscious approach rather than state-of-the-art safety.

Better practice

use the more fail-safe latch and more widely separated hydraulic redundancy

Better follow-through

verify the modification; do not trust the stamp; inspect the critical system yourself

Reflection

When a failure is known to be dangerous, what should happen before the next flight or release: a patch, a redesign, or a grounding of the system?

Fleddermann, Ch. 5, pp. 92–93

Patterns across the cases

Case	Accident type(s)	Main lesson
Hurricane Katrina	systemic + engineered	critical structures need better margins and updated hazard assumptions
Firestone / Explorer	engineered + systemic	test interacting parts, not only isolated parts
Hyatt Regency	engineered + procedural	design changes must be rechecked as engineering work
Ford Crown Victoria	engineered + procedural	legal compliance is not enough when the use context is harsher
Teton Dam	engineered + procedural	one barrier is not enough; leakage and monitoring matter
DC-10	engineered + procedural + systemic	fail-safe choices and thorough inspection are essential

Engagement

Which pattern is most likely to show up in software or platform failures: a bad step, a bad design choice, or a chain of small errors that no one person sees as dangerous?

Chapter 5 Summary

1. Safety is a design duty

Nothing engineers build is perfectly risk free, but designs still have to be as safe as reasonably possible.

2. Risk is judged, not just measured

Voluntariness, reversibility, probability, and timing all change how people perceive risk.

Takeaway

The safest engineering decisions are the ones that reduce the chance that ordinary human error can become a disaster.

3. Safe design climbs in difficulty

Move from legal compliance to accepted practice to alternative designs to foreseeable misuse.

4. Systemic accidents are the hardest

ValuJet shows how many small mistakes can combine into one catastrophic failure.

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